



Safety Data Sheet

acc. to 29 CFR 1910.1200 App D

Drexel Defol® 5 (contains fire retardant)

Version number: 1.1

Date of compilation: 2026-02-16

SECTION 1: Identification

1.1 Product identifier

Trade name **Drexel Defol® 5 (contains fire retardant)**
EPA Reg. No. 19713-388

1.2 Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses Defoliant
Desiccant

1.3 Details of the supplier of the safety data sheet

Drexel Chemical Company, Inc
1700 Channel Avenue
Memphis TN 38113-0327
United States

Telephone: +1 (901) 774-4370
e-mail: info@drexchem.com
Website: https://www.DrexChem.com

e-mail (competent person) cbothwell@drexchem.com (Cyndi Bothwell)

1.4 Emergency telephone number

Emergency information service CHEMTREC 800-424-9300
DREXEL CHEMICAL CO. 901-774-4370

This product is an EPA FIFRA registered pesticide. Some of the classifications on this SDS are not the same as the FIFRA label. Certain sections of this SDS are superseded by federal law governed by EPA for a registered pesticide. Refer to SECTION 15: REGULATORY INFORMATION for explanation.

SECTION 2: Hazard(s) identification

2.1 Classification of the substance or mixture

Classification acc. to OSHA "Hazard Communication Standard" (29 CFR 1910.1200)

Hazard class and category code(s)

Section	Hazard class	Category	Hazard class and category	Hazard statement
3.1I	acute toxicity (inhal.)	4	Acute Tox. 4	H332
3.2	skin corrosion/irritation	2	Skin Irrit. 2	H315
3.3	serious eye damage/eye irritation	2B	Eye Irrit. 2B	H320
4.1A	hazardous to the aquatic environment - acute hazard	3	Aquatic Acute 3	H402
4.1C	hazardous to the aquatic environment - chronic hazard	4	Aquatic Chronic 4	H413

2.2 Label elements

Labelling acc. to OSHA "Hazard Communication Standard" (29 CFR 1910.1200)

- Signal word warning



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- Pictograms

GHS07



Hazard statements

Causes skin and eye irritation.
Harmful if inhaled.
Harmful to aquatic life.
May cause long lasting harmful effects to aquatic life.

- Precautionary statements

Obtain, read and follow all safety instructions before use.
Avoid breathing dust/fume/gas/mist/vapours/spray.
Wash hands, forearms and face thoroughly after handling.
Do not eat, drink or smoke when using this product.
Use only outdoors or with adequate ventilation.
Avoid release to the environment.
Wear protective gloves.
In case of inadequate ventilation wear respiratory protection.
IF ON SKIN: Wash with plenty of water.
IF INHALED: Remove person to fresh air and keep comfortable for breathing.
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
If skin irritation or rash occurs: Get medical help.
If eye irritation persists: Get medical advice/attention.
Take off contaminated clothing and wash it before reuse.
Store in a well-ventilated place. Keep container tightly closed.
Store in a well-ventilated place. Keep cool.
Store locked up.
Store separately.
Dispose of contents/container in accordance with local/regional/national/international regulations.

2.3 Other hazards

Results of PBT and vPvB assessment

Does not contain a PBT-/vPvB-substance at a concentration of $\geq 0.1\%$.

Endocrine disrupting properties

Does not contain an endocrine disruptor (ED) in a concentration of $\geq 0.1\%$.

SECTION 3: Composition/information on ingredients

3.1 Substances

Not relevant (mixture)

3.2 Mixtures



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Description of the mixture

Name of substance	Identifier	Wt%	Classification acc. to GHS	Pictograms
Inert Ingredients	CAS No non-hazardous	57.7		
Sodium Chlorate (Denatured)	CAS No 7775-09-9	42.3	Ox. Sol. 1 / H271 Acute Tox. 4 / H302 Aquatic Chronic 2 / H411	  

The specific chemical identity and/or exact percentage (concentration) of composition has been withheld as a trade secret.

SECTION 4: First-aid measures

4.1 Description of first-aid measures

Following inhalation

Provide fresh air. If breathing is irregular or stopped, immediately seek medical assistance and start first aid actions. In case of respiratory tract irritation, consult a physician.

Following skin contact

After contact with skin, take off immediately all contaminated clothing, and wash immediately with plenty of water. In all cases of doubt, or when symptoms persist, seek medical advice.

Following eye contact

Remove contact lenses, if present and easy to do. Continue rinsing. Irrigate copiously with clean, fresh water for at least 10 minutes, holding the eyelids apart. Get medical advice/attention.

Following ingestion

Call a poison control center or doctor immediately for treatment advice. Rinse mouth with water (only if the person is conscious). Do NOT induce vomiting.

4.2 Most important symptoms and effects, both acute and delayed

Symptoms and effects are not known to date.

4.3 Indication of any immediate medical attention and special treatment needed

If medical advice is needed, have product container or label at hand. This product has low oral, dermal, and inhalation toxicity. If swallowed, gastric lavage using an endotracheal tube may be preferred to vomiting. Treat symptomatically. This product is a mild eye irritant.

SECTION 5: Fire-fighting measures

5.1 Extinguishing media

Suitable extinguishing media

Water spray, Water mist, Water fog

Unsuitable extinguishing media

Water jet, Foam, Dry extinguishing powder, Carbon dioxide (CO₂)

5.2 Special hazards arising from the substance or mixture

Non-combustible, substance itself does not burn but may decompose upon heating to produce corrosive fumes. Closed containers may rupture if exposed to excess heat or flame due to a build-up of internal pressure.

Hazardous combustion products

Carbon monoxide (CO), Carbon dioxide (CO₂), Thermal decomposition during a fire can produce toxic fumes and irritating gases.

5.3 Advice for firefighters

Keep containers cool with water spray. Coordinate firefighting measures to the fire surroundings. Do not allow firefighting water to enter drains or water courses. Collect contaminated firefighting water separately. Fight fire with



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normal precautions from a reasonable distance.

Special protective equipment for firefighters

Chemical protective clothing, Wear self-contained breathing apparatus, MSHA/NIOSH, Use full face shield and operate in positive pressure mode., Avoid contact with this material during firefighting operations. If contact is likely, change to full chemical resistant firefighting clothing with self-contained breathing apparatus. If this is not available, wear full chemical resistant clothing with self-contained breathing apparatus and fight fire from a remote location. For protective equipment in post-fire or non-fire clean-up situations, refer to the relevant sections.

SECTION 6: Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

For non-emergency personnel

Remove persons to safety.

For emergency responders

Isolate area. Ventilate area of leak or spill. Use appropriate safety equipment.

6.2 Environmental precautions

Do not flush into surface water or sanitary sewer system. Prevent from entering soil, ditches, sewers, waterways, and/or groundwater.

6.3 Methods and material for containment and cleaning up

Advice on how to contain a spill

Covering of drains, Stop the flow of material, if this is without risk.

Advice on how to clean up a spill

absorbent material (e.g. sand, diatomaceous earth, acid binder, universal binder, sawdust, etc.), Sweep up and shovel into suitable containers for disposal. Collect in suitable and properly labeled containers. Large spills: Contact Drexel Chemical Company for clean-up assistance.

6.4 Reference to other sections

Refer to Section 7: HANDLING AND STORAGE, for additional precautionary measures. Personal protective equipment: see section 8. Refer to SECTION 12: Ecological Information. Disposal considerations: see section 13.

SECTION 7: Handling and storage

7.1 Precautions for safe handling

Recommendations

- Measures to prevent fire as well as aerosol and dust generation

Use only in well-ventilated areas. Prevent from heating up above 50 °C/122 °F. Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.

Advice on general occupational hygiene

Wash hands after use. Do not eat, drink and smoke in work areas. Keep away from food, drink and animal feedingstuffs. Avoid contact with eyes, skin, and clothing. Avoid breathing dust/fume/gas/mist/vapors/spray. Wear chemical protective equipment when handling. Wear long-sleeved shirt, long pants, and shoes with socks when handling.

7.2 Conditions for safe storage, including any incompatibilities

Managing of associated risks

- Explosive atmospheres

Keep container tightly closed and in a well-ventilated place. Store at temperatures not exceeding 50 °C/122 °F. Keep cool.

- Packaging compatibilities

Keep only in original container.



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SECTION 8: Exposure controls/personal protection

8.1 Control parameters

Occupational exposure limit values (Workplace Exposure Limits)

Country	Name of agent	CAS No	Identifier	TWA [ppm]	TWA [mg/m³]	STEL [ppm]	STEL [mg/m³]	Ceiling-C [ppm]	Ceiling-C [mg/m³]	Notation	Source
US	Sodium chlorate		TLV®		5						ACGIH

Notation

Ceiling-C ceiling value is a limit value above which exposure should not occur

STEL short-term exposure limit: a limit value above which exposure should not occur and which is related to a 15-minute period (unless otherwise specified)

TWA time-weighted average (long-term exposure limit): measured or calculated in relation to a reference period of 8 hours time-weighted average (unless otherwise specified)

8.2 Exposure controls

Appropriate engineering controls

Exhaust ventilation. Ventilate all transport vehicles prior to unloading. Use engineering controls to maintain airborne level below exposure limit requirements or guidelines. If there are no applicable exposure limit requirements or guidelines, use only with adequate ventilation. Local exhaust ventilation may be necessary for some operations. Ensure that eyewash stations and safety showers are close to the workstation location.

Individual protection measures (personal protective equipment)

Eye/face protection

Work with safety glasses. Use safety goggle with side protection. Wear face-shield. Use protective eyewear to guard against splash of liquids.

Skin protection

Use protective clothing chemically resistant to this material. Selection of specific items such as face shield, boots, apron, or full body suit will depend on the task. Safety shower should be located in immediate work area. Remove contaminated clothing immediately, wash skin area with soap and water, and launder clothing before reuse or dispose of properly. Items which cannot be decontaminated, such as shoes, belts and watchbands, should be removed and disposed of properly.

- Hand protection

Chemical protection gloves are suitable, which are tested according to EN 374. Examples of preferred glove barrier materials include Neoprene, Nitrile/butadiene rubber or Polyvinyl chloride. The selection of gloves for a particular application and duration of use in the workplace should also be taken into account all relevant workplace factors such as, but not limited to: other chemicals which may be handled, physical requirements (cut/ puncture protection, dexterity, thermal protection), potential body reactions to gloves materials, as well as the instructions / specs provided by the supplier of gloves. .

- Other protection measures

Avoid ingestion of even very small amounts; do not consume or store food or tobacco in the work area; wash hands and face thoroughly with soap and water before smoking or eating. Avoid getting wash water in eyes.

Respiratory protection

Respiratory protection should be worn when there is a potential to exceed the exposure limit requirements or guidelines. If there are no applicable exposure limit requirements or guidelines, use an approved respirator. Selection of air-purifying or positive-pressure supplied-air will depend on the specific operation and the potential airborne concentration of the material. When handling in enclosed areas, when large quantities of dusts are generated or prolonged exposure is possible in excess of the TLV, use a respirator with either an organic vapor-removing cartridge with a prefilter approved for pesticides (MSHA/NIOSH approval number prefix TC-23C) or a canister approved for pesticides (MSHA/NIOSH approval number prefix TC-14G).



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SECTION 9: Physical and chemical properties

9.1 Information on basic physical and chemical properties

Physical state	liquid
Color	colorless
Odor	odorless
Melting point/freezing point	<32 °F
Boiling point or initial boiling point and boiling range	>212 °F
Flammability	non-combustible
Lower and upper explosion limit	not determined
Flash point	>200 °F
Auto-ignition temperature	not determined
Decomposition temperature	not relevant
pH (value)	8 – 9
Dynamic viscosity	7.4 cP

Solubility(ies)

Water solubility	miscible in any proportion
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Partition coefficient

Partition coefficient n-octanol/water (log value)	this information is not available
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Density and/or relative density

Density	11.67 – 11.77 lb/gal at 68 °F
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SECTION 10: Stability and reactivity

10.1 Reactivity

This material is not reactive under normal ambient conditions.

10.2 Chemical stability

Stable under normal conditions of use.

10.3 Possibility of hazardous reactions

No known hazardous reactions.

10.4 Conditions to avoid



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Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. Avoid high temperatures above 130°F (54.4°C).

10.5 Incompatible materials

Reducing agents, If concentrate solution dries on flammable objects can act as an oxidizer.

10.6 Hazardous decomposition products

Hazardous combustion products: see section 5.

SECTION 11: Toxicological information

11.1 Information on toxicological effects

Classification procedure

The method for classification of the mixture is based on ingredients of the mixture (additivity formula).

Classification acc. to OSHA "Hazard Communication Standard" (29 CFR 1910.1200)

Acute toxicity

May be harmful in contact with skin. Harmful if inhaled.

- Values for lethality

Exposure route	Endpoint	Value	Species	Notes
oral	LD50	>5,000 mg/kg	rat	
dermal	LD50	>5,000 mg/kg	rat	
inhalation: vapor	LC50	2.09 mg/l/1h	rat	

- Acute toxicity of components

Name of substance	CAS No	Exposure route	ATE
Sodium Chlorate	7775-09-9	oral	1,200 mg/kg
Sodium Chlorate	7775-09-9	dermal	>2,000 mg/kg
Sodium Chlorate	7775-09-9	inhalation: vapor	>5.59 mg/l/4h

Skin corrosion/irritation

Causes mild skin irritation.

Serious eye damage/eye irritation

Causes eye irritation.

Respiratory or skin sensitization

Shall not be classified as a respiratory or skin sensitizer.

Germ cell mutagenicity

Shall not be classified as germ cell mutagenic.

Carcinogenicity

Shall not be classified as carcinogenic.

Reproductive toxicity

Shall not be classified as a reproductive toxicant.

Aspiration hazard

Shall not be classified as presenting an aspiration hazard.

Symptoms related to the physical, chemical and toxicological characteristics



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Delayed and immediate effects as well as chronic effects from short and long-term exposure

May cause thyroid damage based on animal data.

SECTION 12: Ecological information

12.1 Toxicity

This product is practically non-toxic to fish, poses no acute risks to birds, and is slightly toxic to aquatic invertebrates and plants.

Aquatic toxicity (acute)

Endpoint	Value	Species	Exposure time
LC50	1,000 mg/l	fish	96 h
EC50	920 mg/kg	daphnia magna	48 h
EC50	43 mg/l	aquatic plants	48 h
EC50	133 mg/l	algae	96 h

Aquatic toxicity (acute) of components

Name of substance	CAS No	Endpoint	Value	Species	Exposure time
Sodium Chlorate	7775-09-9	LC50	13,500 mg/l	fathead minnow (Pimephales promelas)	96 h
Sodium Chlorate	7775-09-9	LC50	1,750 mg/l	rainbow trout (Oncorhynchus mykiss)	96 h

12.2 Persistence and degradability

Low potential to bioaccumulate, is highly soluble and is completely ionized in water.

12.3 Bioaccumulative potential

Low potential to bioaccumulate, is highly soluble and is completely ionized in water.

12.4 Mobility in soil

Data are not available.

12.5 Results of PBT and vPvB assessment

Does not contain a PBT-/vPvB-substance at a concentration of $\geq 0.1\%$.

12.6 Endocrine disrupting properties

Does not contain an endocrine disruptor (ED) in a concentration of $\geq 0.1\%$.

12.7 Other adverse effects

Do not contaminate water supplies, lakes, streams, ponds, or drains with this product.

SECTION 13: Disposal considerations

Remarks

If wastes and/or containers cannot be disposed of according to the product label directions, disposal of this material must be in accordance with your local or area regulatory authorities. This information presented below only applies to the material as supplied. The identification based on characteristic(s) or listing may not apply if the material has been used or otherwise contaminated. It is the responsibility of the waste generator to determine the toxicity and physical properties of the material generated to determine the proper waste identification and disposal methods in compliance with applicable regulations. If the material as supplied becomes a waste, follow all applicable regional, national and local laws.



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SECTION 14: Transport information

14.1	DOT: Not-Regulated IMDG: Not-Regulated IATA/ICAO: Not-Regulated
14.2 UN number	UN 2428
14.3 UN proper shipping name	Sodium Chlorate, Aqueous Solution
14.4 Transport hazard class(es)	Not Applicable
14.5 Packing group	III
14.6 Reportable Quantity	Not Applicable
14.7 Environmental hazards	Not Applicable
14.8 Freight Description	Agricultural Defoliant/Desiccant, liquid, n.o.s.
14.9 ERG Guide No.	171
14.10 Transport Information Note	None
This information is not intended to convey all specific regulatory or operational requirements/information relating to this product. Additional transportation system information can be obtained through an authorized sales or customer service representative. It is the responsibility of the transporting organization to follow all applicable laws, regulations, and rules relating to the transportation of the material.	
14.11 Special precautions for user	There is no additional information.
14.12 Transport in bulk according to IMO instruments	The cargo is not intended to be carried in bulk.

Information for each of the UN Model Regulations

Transport of dangerous goods by road or rail (49 CFR US DOT) - Additional information

Not subject to transport regulations.

International Maritime Dangerous Goods Code (IMDG) - Additional information

Not subject to IMDG.

International Civil Aviation Organization (ICAO-IATA/DGR) - Additional information

Not subject to ICAO-IATA.

SECTION 15: Regulatory information

15.1 Safety, health and environmental regulations specific for the product in question

National regulations (United States)

OSHA Hazard Communication Standard (HCS), 29 CFR 1910.1200

This product contains hazardous components as defined under the criteria of the Federal OSHA Hazardous Communication Standard 29 CFR 1910.1200.

Pesticide Registration

This product is a pesticide registered by the Environmental Protection Agency (EPA) and is subject to certain FIFRA labeling requirements under federal pesticide law. These requirements differ from the classification criteria and hazard information required for safety data sheets (SDS), and for workplace labels of non-pesticide chemicals.

EPA Reg. No. 19713-388



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FIFRA Label Signal Word:

CAUTION

FIFRA Label Information

KEEP OUT OF REACH OF CHILDREN Hazards to Humans and Domestic Animals CAUTION: Harmful if swallowed. Causes moderate eye irritation. Avoid contact with eyes, skin or clothing. Remove and wash contaminated clothing before reuse. Wash thoroughly with soap and water after handling and before eating, drinking, chewing gum, using tobacco, or using the toilet.

Toxic Substance Control Act (TSCA)

all ingredients are listed (ACTIVE) or exempt from listing

Superfund Amendment and Reauthorization Act (SARA TITLE III)

- The List of Extremely Hazardous Substances and Their Threshold Planning Quantities (EPCRA Section 302, 304)
not listed
- Specific Toxic Chemical Listings (EPCRA Section 313)
not listed

Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA)

- List of Hazardous Substances and Reportable Quantities (CERCLA section 102a) (40 CFR 302.4)
not listed

Clean Air Act

not listed

Right to Know Hazardous Substance List

- Cleaning Product Right to Know Act Substance List (CA-RTK)
not listed

California Environmental Protection Agency (Cal/EPA): Proposition 65 - Safe Drinking Water and Toxic Enforcement Act of 1987

not listed

Industry or sector specific available guidance(s)

NPCA-HMIS® III

Hazardous Materials Identification System. American Coatings Association.

HEALTH	/ 1
FLAMMABILITY	2
PHYSICAL HAZARD	0
PERSONAL PROTECTION	-

NFPA® 704

National Fire Protection Association: Standard System for the Identification of the Hazards of Materials for Emergency Response (United States).



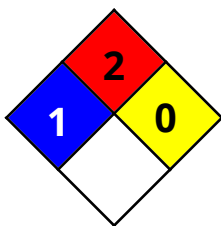
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SECTION 16: Other information, including date of preparation or last revision

Disclaimer

Drexel Chemical Company recommends that each customer or recipient of this SDS to study it carefully and consult appropriate expertise, as necessary or appropriate, to become aware of and understand the data contained in this SDS and any hazards associated with the product. The information herein is provided in good faith and believed to be accurate as of the effective date shown below. However, no warranty, express or implied, is given. Regulatory requirements are subject to change and may differ between various locations. It is the buyer's/user's responsibility to ensure that his activities comply with all federal, state, provincial or local laws. The information presented here pertains only to the product as shipped. Since conditions for use of the product are not under the control of the manufacturer, it is the buyer's/user's duty to determine the conditions necessary for the safe use of this product. Due to the proliferation of sources for information such as manufacturer-specific SDSs, we are not and cannot be responsible for SDSs obtained from any source other than ourselves. If you have obtained an SDS from another source or if you are not sure that the SDS you have is current, please contact us for the most current version.